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ion, although this is far from being firmly established. This center is known to be stable at 77°K in the absence of lithium stabilization. Thus, it is possible that the 380 μ sec lifetime we observe at room temperature is associated with germanium trapped electrons that are not stabilized by lithium before their decay.

The large difference in the time constants observed for the two processes postulated seems somewhat unlikely. Thus, other models of the 3000 Å center should be developed. The present experiment suggests other

experiments, the simplest being the temperature dependence of the lifetimes. It is felt that time resolved spectroscopy of these and other systems will provide a powerful tool for understanding the details of such processes.

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Temperature Dependence of the Energy Gap in GaAs and GaP

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Absorption measurements from 300°–973°K for GaAs and to 1273°K for GaP were made in order to determine the energy gap E_g of these materials at high temperatures. These data, together with previously published results, show that the energy-gap width may be represented by a simple expression for all temperatures. As a function of the absolute temperature T , the direct gap in GaAs is given by $E_g = 1.522 - 5.8 \times 10^{-4} T^2 / (T + 300)$ eV, and the indirect gap in GaP is given by $E_g = 2.338 - 6.2 \times 10^{-4} T^2 / (T + 460)$ eV. The general form of the expression for the energy gap is $E_g = E_g(0) - aT^2 / (T + \beta)$, where $E_g(0)$ is the energy gap at 0°K, β is approximately the 0°K Debye temperature, and a is an empirical constant.

I. INTRODUCTION

In the studies of the solid-state chemistry and physics of semiconductors at high temperatures, it is often necessary to know the width of the energy gap E_g at the temperature under consideration. For example, the intrinsic carrier concentration n_i , which depends exponentially on E_g , can influence both impurity solubility and diffusion. In addition, the energy gap enters many expressions that are often evaluated from near 0°K to slightly in excess of 300°K. An analytical expression for the temperature variation of E_g is very helpful in these calculations. The usual practice for expressing the temperature dependence of E_g has been by a linear variation given as

$$E_g = E_{g0} - bT, \quad (1)$$

where E_{g0} is an energy gap extrapolated to 0°K, b is an empirical constant evaluated near 300°K, and T is the absolute temperature. The purpose of this study was to investigate the adequacy of the linear E_g representation for GaAs and GaP, especially at high temperature.

The determination of the energy-gap temperature dependence is basically a problem of the measurement of the absorption coefficient, and then the assignment of E_g from the absorption data. For GaAs, we have utilized the data of Sturge,¹ and Afromowitz and

Redfield² from 0°K to room temperature, and the measurements described here from room temperature to 973°K to show that the direct energy gap may be expressed as

$$E_g = 1.522 - 5.8 \times 10^{-4} T^2 / (T + 300) \text{ eV}. \quad (2)$$

For GaP, the data of Lorenz, Pettit, and Taylor,³ whose absorption measurements lead to indirect energy-gap values from 6°–868°K, can be expressed as

$$E_g = 2.338 - 6.2 \times 10^{-4} T^2 / (T + 460) \text{ eV}. \quad (3)$$

Our absorption measurements for GaP from 297°–1273°K substantiate the high-temperature results of Ref. 3.

The general form of the energy-gap temperature dependence given by Eq. (2) and (3) is

$$E_g = E_g(0) - aT^2 / (T + \beta), \quad (4)$$

where $E_g(0)$ is the energy gap at 0°K, β is approximately the 0°K Debye temperature θ , and a is an empirical constant. This form of the energy-gap temperature dependence was proposed by Varshni⁴ on the basis of theoretical considerations by Vasileff⁵ and Adams.⁶ Vasileff⁵ considered the shift of the band

² M. A. Afromowitz and D. Redfield, Intern. Conf. on the Phys. of Semicond., Moscow (1968).

³ M. R. Lorenz, G. D. Pettit, and R. C. Taylor, Phys. Rev. **171**, 876 (1968).

⁴ Y. P. Varshni, Physica **34**, 149 (1967).

⁵ H. D. Vasileff, Phys. Rev. **105**, 441 (1957).

⁶ E. N. Adams, Phys. Rev. **107**, 671 (1957).

¹ M. D. Sturge, Phys. Rev. **127**, 768 (1962).

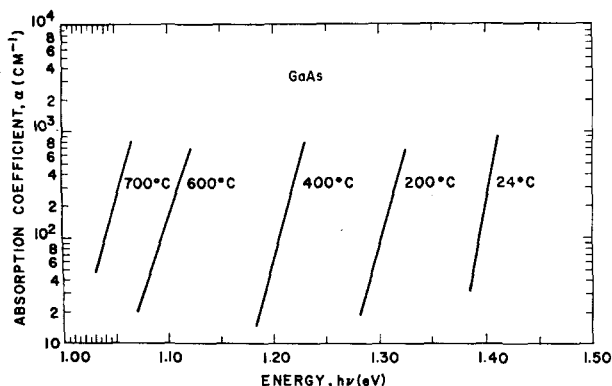


FIG. 1. The absorption coefficient of GaAs from 24°–700°C.

edge by electron-phonon interactions, and this calculation led Varshni⁴ to suggest that E_g should vary linearly with temperature, for $T \gg \theta$, and quadratically with temperature for $T \ll \theta$. Adams⁹ points out that Vasileff's theory⁵ fails for the usual semiconductors, but agrees with Vasileff's conclusions on the temperature dependence of the band edge. The results summarized by Eq. (2) and (3) support the concept that the variation of the energy gap is largely due to electron-phonon interactions rather than thermal expansion.

The two methods which are generally considered for determination of the energy gap in semiconductors are Hall measurements in the intrinsic region and absorption coefficient measurements. Hall measurements are unsuitable for several reasons. For wide energy-gap materials such as GaAs and GaP, the materials are extrinsic at low temperatures, while at high temperatures, electrical contact melting and crystal decomposition are rather formidable problems. In addition, considerable uncertainty results in the theoretical interpretation of the Hall and resistivity measurements in terms of n_i and E_g . The distinct features in the low-temperature absorption spectrum, which are due to free-exciton formation, permit assignment of accurate values of E_g . Above room temperature, the exciton absorption is no longer discernable for either GaAs or GaP. Lorenz *et al.*³ were able to obtain the GaP indirect energy gap from the best fit of expressions from Elliott's theory⁷ for optical absorption in semiconductors to the experimental absorption spectrum. Afromowitz and Redfield² have shown that changes in the direct energy gap for GaAs can be obtained by the shift in energy at a constant value of absorption coefficient. The temperature dependence of the energy gap in GaAs and GaP reported here has been obtained from absorption measurements by these methods.

II. EXPERIMENTAL

The GaAs crystal used in the absorption measurements was "undoped," boat-grown material purchased

⁷ R. J. Elliott, *Phys. Rev.* **108**, 1384 (1957).

from the Monsanto Company. The GaP wafer was grown by the open-tube halogen-transport process⁸ and was also "undoped". Both the GaAs and GaP samples had a net donor concentration of approximately 10^{16} cm^{-3} and were mechanically polished to obtain flat-parallel surfaces. The GaAs sample was 4 mil thick, and the GaP sample was 15 mil thick.

The samples were mounted in an evacuated fused-silica optical cell with approximately 1 mm thick walls with an inside (optical path) dimension of 0.5 mm. Along with each crystal, enough As or P was included in the cell to give a total pressure, when completely vaporized, of a few tenths of an atmosphere. The addition of As or P was necessary to reduce surface damage to the samples at high temperatures. The cells were heated in a small hand-made furnace with multiple windings to aid in the reduction of temperature gradients. Calibration checks indicated that the optical absorption through 0.5 mm of As₄ or P₂ at about one atmosphere was negligible in the spectral regions of interest.

The transmission spectra were determined by the sample-in-sample-out technique. The sample was illuminated with collimated light from a 250 W tungsten-iodine lamp. Cutoff filters were used to isolate the approximate portion of the spectrum being studied, and the light was chopped before entrance to the furnace. After passing through the furnace, the light was focused on the entrance slit of a Jarrell-Ash 0.5 m Ebert scanning spectrometer. The visible or infrared radiation at the exit of the spectrometer was detected with a cooled (S-1 response) photomultiplier. The resulting photomultiplier signal was amplified by a Princeton Applied Research lock-in amplifier system which drives a Leeds and Northrup recorder. The absorption coefficient α was obtained from the ratio T of the transmitted intensity I_t to the incident intensity I_0 by the usual expression that considers multiple internal reflections:

$$T = I_t/I_0 = [(1-R)^2 \exp(-\alpha d) / 1 - R^2 \exp(-2\alpha d)], \quad (5)$$

where R is the reflectivity, and d is the sample thickness.

III. RESULTS AND DISCUSSION

A. GaAs

Absorption coefficients obtained by Eq. (5) from the sample transmittance measurements are shown in Fig. 1. The reflectivity R was taken as 0.30 (Ref. 9 and 10). The rapid decrease in photomultiplier sensitivity below 1.0 eV prevented transmittance meas-

⁸ L. C. Luther and D. D. Roccasecca, *J. Electrochem. Soc.* **115**, 850 (1968).

⁹ D. T. F. Marple, *J. Appl. Phys.* **35**, 1241 (1964).

¹⁰ H. R. Philipp and H. Ehrenreich, *Phys. Rev.* **129**, 1550 (1963).

urements at temperatures in excess of 700°C. Within the uncertainties of the experimental measurements, the slopes of the absorption curves do not exhibit a temperature dependence.

Accurate determination of the energy gap, by fitting the measured absorption coefficient with Elliott's theory,⁷ requires resolving the exciton-absorption peak. Absorption-coefficient values at the exciton peak are in the 10^4 cm^{-1} range for direct-gap materials such as GaAs. Not only are very thin samples necessary for measurement of such large values of α , but the exciton-absorption peak is no longer resolved at room temperature and above. An alternative procedure for assignment of E_g from the absorption coefficient has been demonstrated by Afromowitz and Redfield.² By comparison with Sturge's results,¹ they showed that the change in energy gap with temperature may be assigned from the change in energy at a constant α .

TABLE I. Comparison of the experimental direct energy gap in GaAs with the values obtained from Eqs. (2) and (6).

Temperature (°K)	Experimental E_g (eV)	$E_g = 1.522 - 5.8 \times 10^{-4} T^2 / (T + 300)$ (eV)	$E_g = 1.53 - 5.0 \times 10^{-4} T$ (eV)
21	1.521 (Ref. 1)	1.521	1.520
55	1.518 (Ref. 1)	1.517	1.503
90	1.511 (Ref. 1)	1.510	1.485
185	1.479 (Ref. 1)	1.481	1.438
294	1.435 (Ref. 1)	1.437	1.383
473	1.350	1.354	1.294
673	1.253	1.252	1.194
873	1.147	1.145	1.094
973	1.090	1.091	1.044

Assignment of E_g in this manner requires that the exponential slope of α at energies less than E_g is unaffected by changes in temperature. A constant slope of α occurs for samples in which the number of ionized impurities is temperature independent.

Values of the energy gap were obtained from Fig. 1 by subtracting the change in energy at a constant absorption coefficient of 40 cm^{-1} from Sturge's¹ E_g value of 1.435 eV at 294°K. The uncertainty in the assignment of E_g is about $\pm 0.003 \text{ eV}$ at 473°K and increases to $\pm 0.005 \text{ eV}$ at 973°K. Because the exciton-absorption peak is not resolved at room temperature,^{1,11} Vrehen¹¹ obtained the differential absorption spectrum in an oscillating electric field. On the basis of his results, he concluded that Sturge's value of E_g at 295°K is about 0.007 eV too high. Because Vrehen¹¹ only has E_g values at 2°, 77°, and 295°K, his data are not at sufficiently close temperature intervals for

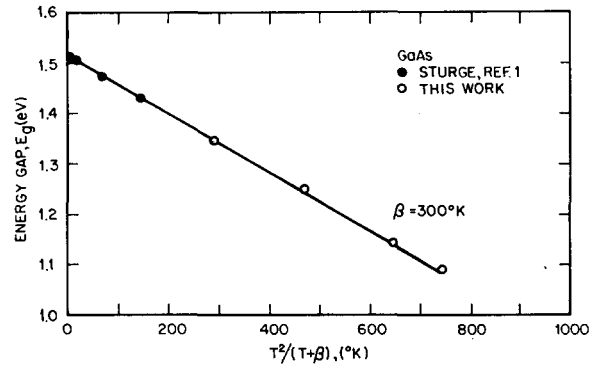


FIG. 2. The energy gap for GaAs vs $T^2/(T+300)$. A straight line was obtained for $\beta = 300^\circ\text{K}$.

determination of a and β in Eq. (4). For this reason we have utilized Sturge's data,¹ but realize that more complete data in the future may require reevaluation of a and β . The resulting E_g values, together with Sturge's values, are summarized in Table I. These E_g values are plotted in Fig. 2 to show that the data give a straight line when plotted as a function of $T^2/(T+300)$, with $\beta = 300^\circ\text{K}$, and obey Eq. (4). The 0°K Debye temperature for GaAs calculated from the elastic constants is 344°K (Ref. 12). Also, from Fig. 2, the value of a is found to be 5.8×10^{-4} , which gives $E_g = 1.522 - 5.8 \times 10^{-4} T^2 / (T + 300)$, as expressed in Eq. (2). Comparison in Table I of the experimental values of E_g with the values calculated from Eq. (2) results in an agreement of $\pm 0.004 \text{ eV}$ over the entire temperature range. This agreement demonstrates that Eq. (2) is useful for expressing E_g at any temperature.

Previously, the temperature dependence of the GaAs energy gap has been given by the linear expression¹³

$$E_g = 1.53 - 5.0 \times 10^{-4} T. \quad (6)$$

Values of E_g calculated from Eq. (6) are given in Table I to demonstrate how poorly this linear approxi-

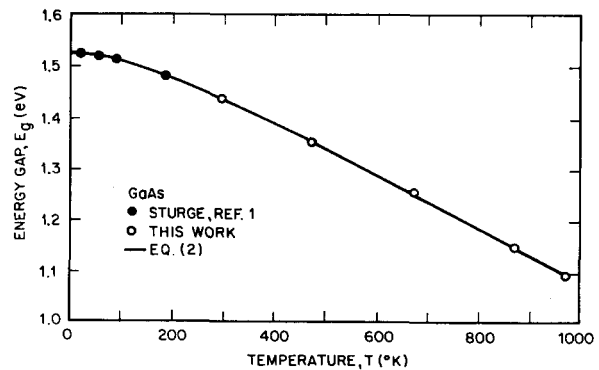


FIG. 3. The energy gap as a function of temperature for GaAs. The solid curve was calculated from $E_g = 1.522 - 5.8 \times 10^{-4} T^2 / (T + 300)$.

¹¹ Q. H. F. Vrehen, J. Phys. Chem. Solids 29, 129 (1968).

¹² E. F. Steigmeier, Appl. Phys. Letters 3, 6 (1963).

¹³ F. Oswald, Z. Naturforsch. 10A, 927 (1955).

TABLE II. Comparison of the experimental indirect energy gap in GaP with the values obtained from Eqs. (3) and (7).

Temperature (°K)	Experimental E_g (eV)		$E_g = 2.338$ -6.2×10^{-4} $T^2/(T+460)$ (eV)	$E_g = 2.365$ $-3.57 \times 10^{-4} T$ (eV)
	This work	Ref. 3		
0		2.328	2.338	2.365
100		2.329	2.327	2.329
150		2.315	2.315	2.311
300	2.261	2.261	2.265	2.258
343		2.243	2.247	2.243
417		2.210	2.215	2.216
466		2.188	2.193	2.198
473	2.182		2.189	2.196
545		2.152	2.155	2.170
620		2.118	2.117	2.143
673	2.087		2.090	2.125
759		2.050	2.045	2.095
868		1.986	1.986	2.055
873	1.981		1.984	2.053
1023	1.859		1.873	1.999
1273	1.738		1.758	1.910

mation fits the present experimental values of E_g . Other values of E_{g0} and b in Eq. (1) can be chosen to obtain a fit over a different temperature range, but the linear expression only fits a limited temperature range. The experimental values of E_g and the curve calculated from Eq. (2) are shown in Fig. 3. This figure shows that the temperature variation of the energy gap only approaches a linear dependence at high temperatures, and the linear E_g representation, based on the behavior of E_g near room temperature, does not adequately represent E_g at high temperatures.

B. GaP

The indirect energy-gap values of GaP used to express E_g in terms of Eq. (4) were taken from the results of Lorenz *et al.*³ Their values are in close agreement with Dean *et al.*'s results^{14,15} from 1.6°–300°K. Lorenz *et al.*,³ determined E_g below 400°K from the exciton absorption, and above 400°K, E_g was assigned from a numerical fit of the absorption spectrum with an expression (see Eq. (2), Ref. 3) from Elliott's theory.⁷ Their E_g values for temperatures from 0°–868°K are summarized in Table II, and have been plotted in Fig. 4 to show the data may be fit by Eq. (4), with $\beta = 460^\circ\text{K}$ and $a = 6.2 \times 10^{-4}$. The indirect

energy gap in GaP has been written as $E_g = 2.338 - 6.2 \times 10^{-4} T^2/(T+460)$ in Eq. (3) and is within ± 0.005 eV of Lorenz *et al.*'s³ experimental values over the entire temperature range. The 0°K Debye temperature for GaP, calculated from the elastic constants, is 446°K (Ref. 12). The E_g values, represented by crosses, are our results obtained from absorption measurements between 300° and 1273°K. Our samples showed an additional energy-dependent absorption of 5–10 cm^{-1} on the low-energy side of the intrinsic absorption edge. The additional low-energy absorption was attributed to nonspecular reflectance due to surface roughness.¹⁶ However, our results (shown in Figs. 4 and 5 and Table II) substantiate the results of Ref. 3, which are consistently greater than the values of E_g reported by Subashiev and Chalikyan.¹⁷ At the highest temperatures, our data has a greater uncertainty because of the increased effect of the surface roughness, and only serves to show that the extension of Eq. (3) above 868°K is reasonable.

Between 140° and 300°K, the variation of E_g has previously been fit with the linear relation¹⁴

$$E_g = 2.365 - 3.57 \times 10^{-4} T. \quad (7)$$

Although Eq. (7) was intended to represent E_g only over the specified temperature range, we have compared it with the experimental values in Table II over a much greater range. The experimental values of E_g and the curve calculated from Eq. (3) are also shown in Fig. 5. Figure 5 and Table II clearly illustrate that the linear expression for E_g , based on values near room

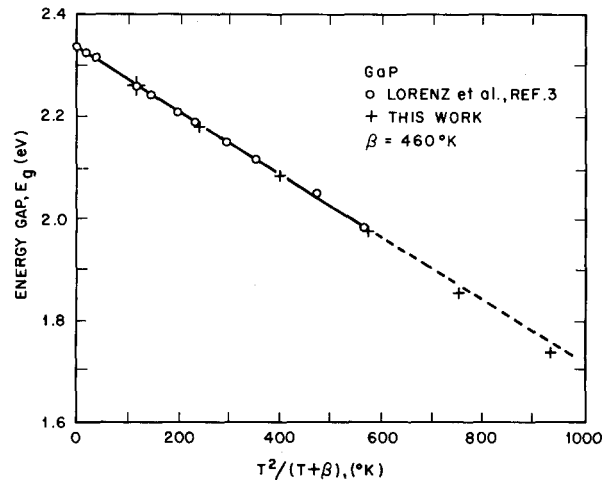


FIG. 4. The energy gap for GaP vs $T^2/(T+\beta)$. The solid straight line was obtained for $\beta = 460^\circ\text{K}$ with the data of Lorenz, Pettit, and Taylor,³ and the dashed line is an extension of this line to higher temperatures.

¹⁶ H. E. Bennett and J. O. Porteus, J. Opt. Soc. Am. **51**, 123 (1961).

¹⁷ V. K. Subashiev and G. A. Chalikyan, Phys. Status Solidi **13**, K91 (1966).

¹⁴ P. J. Dean and D. G. Thomas, Phys. Rev. **150**, 690 (1966).

¹⁵ P. J. Dean, G. Kaminsky, and R. B. Zetterstrom, J. Appl. Phys. **38**, 3551 (1967).

temperature, is not an adequate representation for GaP at high temperatures.

C. General Discussion

The results presented in Parts A and B of this section demonstrated that a linear representation of the temperature dependence of the energy gap is inadequate for GaAs and GaP. The linear variation of the energy gap with temperature, as represented by Eq. (1), was originally based upon the assignment of thermal expansion as the principal mechanism for the energy-gap temperature dependence. The temperature coefficient term b had been taken as proportional to the coefficient of linear expansion.^{18,19} Actually, the coefficient of linear expansion has a strong temperature dependence and even becomes negative at low temperature.²⁰ DeMeis²¹ calculated the change in the energy gap for GaAs by including the temperature dependence of the coefficient of linear expansion. He found that the thermal-expansion contribution to the variation of E_g was approximately 10% or less up to 300°K. Since the contribution due to lattice expansion is small, and the experimental temperature dependence has the form predicted for electron-phonon interactions,⁵ it appears that the electron-phonon interaction is the dominant mechanism for the energy-gap temperature dependence in GaAs. The dominance of the electron-phonon interactions has been demonstrated for Ge and Si by Cohen,²² and for diamond by Dean and Jones.²³ Similarly, it appears reasonable to consider the electron-phonon interaction the dominant mechanism for the temperature variation of E_g in GaP.

The fit of the experimental E_g to $E_g = E_g(0) - aT^2/(T+\beta)$ of Eq. (4) only leads to unique values of a and β when the temperature range extends from near 0°K to temperatures larger than the 0°K Debye temperature. The E_g value near 0°K establishes $E_g(0)$, while several combinations of a and β will lead to comparable agreement of Eq. (4) with the experimental E_g up to room temperature. For example, Varshni⁴ applied Eq. (4) to Sturge's GaAs data¹ with the assignment of $a=8.87 \times 10^{-4}$, and $\beta=572$. The E_g calculated with these values is within 0.002 eV of Sturge's 21°–294°K E_g values, but gives an E_g value that is about 0.10 eV too small at 973°K. Our values of $a=5.8 \times 10^{-4}$ and $\beta=300$ give E_g values that agree equally well with Sturge's results, but also agree closely at the high temperatures. No other values of a

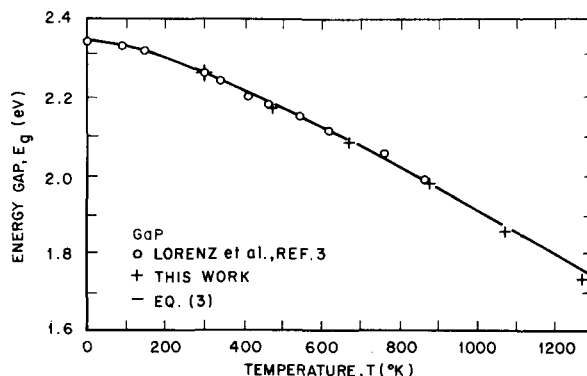


FIG. 5. The energy gap as a function of temperature for GaP. The solid curve was calculated from $E_g = 2.338 - 6.2 \times 10^{-4} T^2/(T+460)$.

and β fit the entire temperature range as well as these values. We have considered a and β for other semiconductors in which the experimental variation of the energy gap has been correctly evaluated by observation of the free-exiton formation. Unique values could not be obtained because of the lack of high-temperature experimental data for E_g . Values of a and β significantly different from the ones assigned by Varshni⁴ for Si, Ge, and InP give as good a fit to the experimental E_g as those he reported. Therefore, in order to assign unique values for a and β , high-temperature E_g data are necessary.

IV. CONCLUSIONS

The form of the energy-gap temperature dependence that was suggested by Varshni,⁴

$$E_g = E_g(0) - aT^2/(T+\beta),$$

has been found to represent the experimental data for GaAs and GaP over an extensive temperature range. In this expression, $E_g(0)$ is the energy gap at 0°K, a is an empirical constant that is approximately 6×10^{-4} , and β is approximately the 0°K Debye temperature. The direct gap in GaAs was found to be $E_g = 1.522 - 5.8 \times 10^{-4} T^2/(T+300)$, while the indirect gap in GaP is $E_g = 2.338 - 6.2 \times 10^{-4} T^2/(T+460)$.

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¹⁸ J. Bardeen and W. Shockley, Phys. Rev. **80**, 72 (1950).

¹⁹ H. Brooks, *Advances in Electronics and Electron Physics*, L. Marton, Ed. (Academic Press, Inc., New York, 1955), Vol. 7, p. 121.

²⁰ S. I. Novikova, Sov. Phys.—Solid State **3**, 129 (1961).

²¹ W. M. DeMeis, Tech. Rept. No. ARPA-16, Gordon McKay Lab., Harvard University, Cambridge, Mass. (1965).

²² M. L. Cohen, Phys. Rev. **128**, 131 (1962).

²³ P. J. Dean and I. H. Jones, Phys. Rev. **133**, A1698 (1964).